

Innovation Project — Team Packets

▪ Set Two

FIRST® LEGO® League Challenge · BIOGLOW (2026–27) · Grades 3–6 · 30 fresh full packets (Forest / Urban / Rural)

□ Forest ×10

□ Urban ×10

□ Rural ×10

Different ideas from Set One

Every packet restores **one broken vital connection** in an ecosystem — the BIOGLOW “canopy” idea, worked in three settings. These are **starters**: the Innovation Project must be the team’s own original work, judged on the official rubric. Each packet runs the five steps — Identify → Interview → Create → Iterate → Communicate.

Forest — 10

- 1 Night Crossing for Spotted Salamanders
- 2 Fuel Stop for Tired Travelers
- 3 Homes for Ground-Nesting Bees
- 4 Clear Skies for Hunting Bats
- 5 A Woodland Water Dish
- 6 High-Rise for Hungry Hawks
- 7 The Leaf-Litter Recycling Crew
- 8 Bring Back the Fireflies' Glow
- 9 Toad Abodes
- 10 A Ladder for Migrating Fish

Urban — 10

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- 13 A Meadow on the Roof
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- 19 Stop the Salt from Reaching the Creek
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Rural — 10

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- 23 Perches for the Farm's Mouse Patrol
- 24 The Dung Beetle Cleanup Crew
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- 27 Early Warning for Invaders
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- 30 A Beetle Bank in the Field

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1 Night Crossing for Spotted Salamanders

Forest

Problem Spotted salamanders can't reach their vernal breeding pools because a woodland road cuts between the forest and the water. · **Solution** A kid-designed low tunnel and drift-fence "salamander crossing" that guides them safely under the road on rainy spring nights.

Mission M8 Tangled, M10 Fragile Microhabitats · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A cardboard-box tunnel with a sloped funnel fence made of folded paper, showing the safe path.

1 • Identify

Every spring, spotted salamanders wake up and march to the same small pond to lay eggs. When a road sits in the way, cars squash them and they never make it. If the grown-ups can't cross, there are no baby salamanders next year.

Research questions

1. Why do salamanders only move on warm, rainy nights?
2. What does a vernal pool have that a stream does not?
3. How far will a salamander travel to reach its breeding pool?

2 • Interview an expert

Who: A herpetologist (amphibian scientist), a park wetland ranger, a county road engineer

Ask them

1. How do you count salamanders without hurting them?
2. What size tunnel do salamanders actually use?
3. How do fences guide the animals toward the tunnel?
4. What time of year is the crossing busiest?
5. Have wildlife tunnels worked in other places?

3 • Create

Materials

- 2 shoeboxes or a cardboard tube
- Construction paper for drift fences
- Tape and a stapler
- A spray bottle of water (rain)
- Small clay or eraser "salamanders"
- A flashlight

Build steps

1. Cut a tube long enough to reach under a pretend road.
2. Fold paper into two angled fences that funnel toward the tunnel mouth.
3. Tape the fences so no gaps let salamanders wander onto the road.
4. Set the pool on one side and the forest on the other.
5. Label the parts: forest, drift fence, tunnel, road, vernal pool.

4 • Iterate & 5 • Communicate

Slide clay salamanders along the fence and see how many find the tunnel instead of the road. Change the fence angle until almost all make it.

Measure: The percent of salamanders that reach the pool without crossing the road.

"It's a rainy March night, and a thousand tiny travelers are trying to cross a river of headlights. Tonight, we build them a bridge underneath."

2 Fuel Stop for Tired Travelers

Forest

Problem Migrating songbirds land in the forest exhausted, but too few berry bushes are left to refuel them before the next leg of their journey. · **Solution** A kid-planned "berry pit stop" that plants native fruiting shrubs in a gap so birds can eat and rest.

Mission M4 Lucky Leaves, M14 Seeds of Renewal · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A tabletop forest gap with sponge "shrubs" and bead "berries" the birds can visit.

1 • Identify

Birds like thrushes and warblers fly hundreds of miles and stop in forests to eat. Fatty native berries are their high-energy fuel. When those bushes disappear, hungry birds can't make it to their winter home.

Research questions

1. Which native berries give birds the most energy?
2. Why can't birds just eat any berry they find?
3. When during the year do migrating birds pass through?

2 • Interview an expert

Who: An ornithologist (bird scientist), a native-plant nursery grower, a bird-banding volunteer

Ask them

1. How do you know a bird is migrating and not local?
2. Which shrubs do birds visit most in fall?
3. How much does a bird eat at one stopover?
4. Do invasive berries help or hurt migrating birds?
5. How long does a bird rest before flying again?

3 • Create

Materials

- A shallow box for the forest floor
- Green sponges cut as shrubs
- Red and purple beads for berries
- Craft-stick trees
- Cotton "birds"
- Glue and markers

Build steps

1. Build a bare gap in the middle of your forest model.
2. Place sponge shrubs in the gap and stick on bead berries.
3. Add taller trees around the edge for cover.
4. Move cotton birds from shrub to shrub, "eating" berries.
5. Label the parts: gap, native shrubs, berries, cover trees.

4 • Iterate & 5 • Communicate

Try clustering shrubs versus spreading them out, and see which layout lets more birds feed safely. Add cover and watch if birds "stay" longer.

Measure: How many berries a model bird can reach in 30 seconds.

"Imagine flying nonstop for three days, then landing in a forest with an empty refrigerator. We're stocking the shelves for one very hungry traveler."

3 Homes for Ground-Nesting Bees

Forest

Problem Most wild bees nest in bare, sunny soil, but leaves, mulch, and mowing cover every open patch, so they have nowhere to dig. · **Solution** Kids build a “bee bank” — a small mound of bare, sandy soil in a sunny spot where ground-nesting bees can dig their tunnels.

Mission M4 Lucky Leaves, M14 Seeds of Renewal · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A tray “bee bank” of sand and soil with pencil-hole “nest tunnels” and a sunny-side label.

1 • Identify

About 7 in 10 wild bees don't live in hives — each female digs her own tunnel in warm, bare soil and stocks it with pollen. When people cover every open patch with mulch, grass, or leaves, these gentle bees can't nest, and fewer flowers get pollinated.

Research questions

1. Which local bees nest in the ground instead of hives?
2. What kind of soil and sun do they need to dig?
3. Where is there bare, sunny ground near our school?

2 • Interview an expert

Who: A native-bee researcher, a master gardener, a county extension agent

Ask them

1. How can I tell a ground-nest hole from an ant hole?
2. What soil do ground-nesting bees like best?
3. How much sun does a bee bank need?
4. Will these bees sting kids?
5. What's the biggest mistake people make trying to help bees?

3 • Create

Materials

- Shallow tray or box lid
- Sand
- Loose soil / dirt
- A pencil (to poke tunnels)
- Small paper bees
- A sunny-spot label / arrow

Build steps

1. Mix sand and soil and mound it in the tray.
2. Smooth a bare, open surface on top.
3. Poke pencil “tunnels” angled into the mound.
4. Add paper bees at the tunnel mouths.
5. Label the parts: bare soil, tunnel, pollen store, sunny side.

4 • Iterate & 5 • Communicate

Tilt the tray toward a lamp “sun” and see which side stays warm and dry; keep the bank where it drains and warms fastest.

Measure: How many pencil-tunnels stay open (don't collapse) in dry vs. damp sand.

“Everyone pictures a beehive — but seven of every ten bees live alone, underground. We built them a home you'd never notice.”

Problem Forest bats can't catch enough insects at night because cluttered, tangled edges and bright lights scramble the echoes they hunt with. · **Solution** A kid-designed "clean flyway" and bat box plan that opens quiet, dark hunting lanes between trees.

Mission M3 Flip the Rock, M10 Fragile Microhabitats · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A dark tunnel maze where a "bat" using sound-bounces finds paper moths.

1 • Identify

Bats "see" with sound by bouncing squeaks off insects in the dark. A single bat eats thousands of bugs a night. Too many tangled branches and glaring lights confuse the echoes, so bats go hungry and bugs take over.

Research questions

1. How does echolocation help a bat catch a moth?
2. Why do bright lights make hunting harder for bats?
3. What insects do forest bats eat the most?

2 • Interview an expert

Who: A bat biologist, a wildlife acoustics researcher, a park naturalist

Ask them

1. How do you record sounds bats make that we can't hear?
2. Why do bats prefer dark flyways?
3. How many insects can one bat eat in a night?
4. What makes a good bat box?
5. How do lights change where bats will fly?

3 • Create

Materials

- A shoebox for a dark flyway
- Craft-stick trees as clutter
- Paper moths
- A marble or ball as the "bat"
- A small flashlight
- Tape and black paper

Build steps

1. Line the box with black paper to make it dark.
2. Place trees to make a cluttered lane and an open lane.
3. Scatter paper moths in both lanes.
4. Roll the "bat" and count moths reached in each lane.
5. Label the parts: bat, open flyway, clutter, moths, bat box.

4 • Iterate & 5 • Communicate

Remove some clutter or turn off the light and test whether the bat catches more moths. Keep the changes that help most.

Measure: Moths reached in the open lane versus the cluttered lane.

"A bat can catch a mosquito in total darkness using nothing but an echo. So why is it going hungry? Someone left the lights on."

Problem In dry spells, small forest animals and insects can't find safe drinking water, so they weaken or leave the area.

• **Solution** Kids design shallow "wildlife watering dishes" with stepping-stones, refilled during droughts, so bees, birds, and small mammals can drink safely.

Mission M9 Research Platform, M5 Reaching Roots • **Connects** BIOBUZZ ★, BIOCORE • **Prototype** A shallow dish with pebble "landing stones" and a slow-drip refill bottle, labeled for who drinks where.

1 • Identify

Animals need water as much as food. In a dry summer, forest streams and puddles disappear, and tiny creatures like bees can drown in steep-sided water. A shallow dish with stones gives them a safe place to sip.

Research questions

1. Which animals struggle to find water in a drought?
2. Why do bees and butterflies drown in deep water?
3. How can the water stay clean and not spread disease?

2 • Interview an expert

Who: A wildlife rehabilitator, a park naturalist, a beekeeper

Ask them

1. Which animals visit water sources the most?
2. How deep is too deep for an insect?
3. How often should a dish be cleaned?
4. Where should it go so predators can't ambush drinkers?
5. Can the wrong setup spread sickness?

3 • Create

Materials

- Shallow dish or plate
- Pebbles / marbles (landing stones)
- A small bottle (drip refill)
- Water
- Paper animals
- A shady-spot label

Build steps

1. Fill the dish with a thin layer of water.
2. Add pebbles so their tops poke above the surface.
3. Set the bottle to drip and keep it topped up.
4. Place paper drinkers on the stones.
5. Label: shallow edge, landing stone, shade, refill.

4 • Iterate & 5 • Communicate

Try different stone heights and water depths; find the setup where a marble "bee" can reach water without going under.

Measure: The water depth at which paper/marble insects can still drink without "drowning."

"A bee can die of thirst three feet from water — if that water is too deep. Watch how one dish and a handful of stones change everything."

Problem Big raptors like ospreys and red-shouldered hawks have nowhere to nest because old, tall dead trees have been cleared from the forest. · **Solution** A kid-designed nest platform that gives raptors a safe, high place to raise chicks and keep rodent numbers in balance.

Mission M12 Forest Elder, M9 Research Platform · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A tall craft-stick tower topped with a paper-plate nest that must hold "eggs" steady in wind.

1 • Identify

Hawks and ospreys need tall, sturdy perches to build big nests and watch for prey. Old dead trees called snags are perfect, but people often cut them down. Without nests, fewer raptors hunt the mice and voles that can overrun a forest.

Research questions

1. Why do raptors need such tall nesting spots?
2. What happens to a forest when top predators disappear?
3. Why are dead trees valuable to wildlife?

2 • Interview an expert

Who: A raptor biologist, a wildlife rehabber, a park forester

Ask them

1. How high should a nest platform be?
2. How do you keep a platform from blowing over?
3. How many mice does a hawk family eat?
4. When do raptors lay their eggs?
5. How do you know if a platform is being used?

3 • Create

Materials

- Craft sticks and a cardboard base
- A paper plate for the nest
- Clay "eggs"
- Straws for support
- A fan or your breath for wind
- Tape and glue

Build steps

1. Build a tall, sturdy tower on a wide base.
2. Fix a paper-plate nest flat on top.
3. Place clay eggs in the nest.
4. Blow wind at it and check if the eggs stay put.
5. Label the parts: platform, nest, eggs, support legs, perch.

4 • Iterate & 5 • Communicate

Widen the base or add cross-braces until the tower survives strong wind without dropping eggs. Test each change the same way.

Measure: The wind strength the tower survives with eggs still in the nest.

"Somewhere a hawk is circling, looking for the tallest tree in the forest to raise her chicks — and finding only stumps. We're building her a penthouse."

Problem When people rake away every fallen leaf, the tiny decomposers that turn leaves into new soil lose their homes and winter shelter. · **Solution** A kid-designed "leave the leaves" plan and a decomposer shelter that keeps the recycling crew fed and cozy.

Mission M3 Flip the Rock, M6 Leafcutter Frenzy · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** Two trays of soil, one with leaf litter, comparing how fast scraps break down.

1 • Identify

Fallen leaves are not trash — they are a blanket where worms, beetles, and fungi turn old leaves into fresh soil and shelter through winter. When every leaf is bagged and hauled away, the decomposers vanish and the soil gets poor.

Research questions

1. Who are the main decomposers in leaf litter?
2. How does leaf litter protect insects in winter?
3. What happens to soil when leaves are removed every year?

2 • Interview an expert

Who: A soil ecologist, a composting educator, a park naturalist

Ask them

1. How long does a leaf take to become soil?
2. What lives in the leaf layer over winter?
3. How does leaf litter help next year's plants?
4. How much litter should we leave behind?
5. What creatures show a healthy soil?

3 • Create

Materials

- Two trays or tubs
- Soil
- Dry leaves
- Small food scraps or paper bits
- A spray bottle
- Craft "worms" and "beetles"

Build steps

1. Fill both trays with the same soil and scraps.
2. Cover one tray with a leaf-litter blanket.
3. Keep both a little damp and wait several days.
4. Compare how fast scraps break down and stay moist.
5. Label the parts: soil, leaf blanket, decomposers, food scraps, new soil.

4 • Iterate & 5 • Communicate

Try different leaf-blanket thicknesses and see which keeps soil moist and breaks scraps down fastest. Keep the best amount.

Measure: How much the scraps shrink after one week under litter.

"Every autumn, the forest drops millions of leaves — and a hidden cleanup crew turns them into next spring's flowers. Please stop taking away their groceries."

Problem Fireflies are blinking out of the forest because bright lights and mowed, dry ground erase the dark, damp habitat they need to signal and mate. · **Solution** A kid-designed "dark-and-damp firefly patch" that keeps the ground moist, unmowed, and shielded from light so their glow can connect them again.

Mission M10 Fragile Microhabitats, M14 Seeds of Renewal · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A dark box where glow-dot "fireflies" are easy or hard to see depending on the light.

1 • Identify

Fireflies flash light to find each other and mate. Their glow is a secret code that only works in darkness. When lights shine all night and grass is mowed short and dry, fireflies can't see each other's signals and disappear.

Research questions

1. Why do fireflies flash their lights?
2. How does bright light hide a firefly's glow?
3. Why do firefly babies need damp ground?

2 • Interview an expert

Who: An entomologist (insect scientist), a firefly-watch volunteer, a dark-sky educator

Ask them

1. Why do different fireflies flash different patterns?
2. What do firefly larvae eat in the soil?
3. How does outdoor lighting hurt fireflies?
4. What kind of habitat brings fireflies back?
5. When is the best time of year to see them?

3 • Create

Materials

- A shoebox with a lid
- Glow-in-the-dark dots or stickers
- Black paper
- A small flashlight
- Damp sponge for moist ground
- Tape

Build steps

1. Line the box dark and add glow-dot fireflies.
2. Add a damp sponge patch for firefly babies.
3. View the glow with the lid closed (dark).
4. Shine the flashlight in and see how the glow disappears.
5. Label the parts: dark patch, fireflies, flash signal, damp ground, stray light.

4 • Iterate & 5 • Communicate

Add a simple shield around the light and see if fireflies become visible again. Try different shields and keep the best.

Measure: How many glow-dots you can see with the light on versus shielded.

"On a summer night, a forest can light up with a thousand tiny lanterns sending love letters in code. One porch light can erase every word."

Problem Toads eat thousands of pest insects, but tidy yards and forests remove the cool, damp hiding spots they need to survive hot days. · **Solution** Kids build “toad abodes” — shady, damp shelters from flowerpots and logs — giving toads a cool place to rest so they stay and hunt pests.

Mission M10 Fragile Microhabitats, M12 Forest Elder · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** An overturned-pot toad house set in damp moss with a doorway and a shady-spot label.

1 • Identify

One toad can eat 10,000 bugs in a summer, helping forests and gardens without any pesticide. But toads dry out fast in the sun, so they need cool, moist hideouts. Remove those, and the toads — and their free pest control — disappear.

Research questions

1. What do toads need to stay cool and wet?
2. How many pest insects can one toad eat?
3. What removes toad hiding spots near us?

2 • Interview an expert

Who: A herpetologist, a gardener, a park naturalist

Ask them

1. Where do toads hide during a hot day?
2. What makes a good toad shelter?
3. Do toads return to the same spot?
4. How do pesticides hurt toads?
5. What’s one easy way a school could help toads?

3 • Create

Materials

- A clay flowerpot (the “abode”)
- Moss or damp soil
- A small water dish
- Twigs / leaves (cover)
- A paper or clay toad
- A shade label

Build steps

1. Chip or tilt the pot to make a doorway.
2. Set it upside down in a shady, damp spot.
3. Tuck moss and a water dish nearby.
4. Add the toad inside the doorway.
5. Label: doorway, shade, damp floor, water.

4 • Iterate & 5 • Communicate

Put the abode in sun vs. shade and feel which stays cooler and damper; keep it where it holds moisture longest.

Measure: How much cooler/damper it is inside the shaded abode than in the open (touch test or a simple thermometer).

“Meet nature’s free exterminator — it eats 10,000 bugs a summer and asks for one thing: a cool, dark place to nap.”

Problem A blocked or too-steep road culvert stops forest-stream fish from swimming upstream to spawn, cutting the water's life in two. · **Solution** A kid-designed "fish ladder" of gentle steps and resting pools that lets fish climb past the barrier.

Mission M8 Tangled, M9 Research Platform · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A ramp with cardboard step-pools that let a "fish" marble climb instead of getting flushed back.

1 • Identify

Many fish swim upstream to lay eggs in the same places every year. When a road pipe is too high or the water inside it rushes too fast, the fish hit a wall and can't get through. Upstream, the eggs never come.

Research questions

1. Why do fish swim upstream to spawn?
2. What makes a culvert impossible for fish to pass?
3. How does a fish ladder help fish climb?

2 • Interview an expert

Who: A fisheries biologist, a stream restoration engineer, a road-crossing inspector

Ask them

1. How high can a fish actually jump?
2. Why do resting pools matter in a fish ladder?
3. How do you measure if fish are getting through?
4. Which fish need to migrate in our streams?
5. What makes a culvert fish-friendly?

3 • Create

Materials

- A cardboard ramp
- Small boxes or caps for step-pools
- A marble or bead "fish"
- A cup of water
- Tape and scissors
- A book to prop the ramp

Build steps

1. Prop the ramp to act as a steep culvert.
2. Add cardboard step-pools down the slope.
3. Pour a little water so it spills pool to pool.
4. Move the "fish" up the steps against the flow.
5. Label the parts: culvert, steps, resting pools, upstream, spawning ground.

4 • Iterate & 5 • Communicate

Change the step height and pool size so the "fish" can climb without being flushed backward. Keep the gentlest ladder that still works.

Measure: How many steps the "fish" can climb before losing to the current.

"A little fish just swam a hundred miles to reach the exact stream where it was born — and slammed into a pipe. We're building the staircase it needs."

Problem Migrating birds crash into shiny city windows because the glass reflects the sky and trees, tricking them into flying straight at it. · **Solution** A kid-designed pattern of dots or cords on the glass that birds can see and avoid.

Mission M1 Drone Survey, M10 Fragile Microhabitats · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** Two "windows" — plain and patterned — tested with a paper bird to see which it avoids.

1 • Identify

Glass buildings act like mirrors, showing sky and trees that aren't really there. Birds fly toward the reflection and hit the window. Millions die this way each year, especially during migration when cities are full of travelers.

Research questions

1. Why can't birds tell glass from open sky?
2. When of year are window crashes worst?
3. What kinds of markings help birds see glass?

2 • Interview an expert

Who: An ornithologist, a building manager, a bird-safe-glass designer

Ask them

1. How do you count birds that hit windows?
2. How far apart should dots be to work?
3. Which windows are the most dangerous?
4. Do markings still let people see out?
5. What time of day are crashes most common?

3 • Create

Materials

- Two clear plastic sheets or lids
- White paint dots or stickers
- A paper "bird"
- A sky/tree picture to reflect
- Tape
- A ruler for spacing

Build steps

1. Prop both plastic sheets up as windows.
2. Leave one plain; add evenly spaced dots to the other.
3. Place a sky picture behind to make reflections.
4. "Fly" the paper bird and note which window it would hit.
5. Label the parts: glass, reflection, dot pattern, safe zone, bird.

4 • Iterate & 5 • Communicate

Try different dot spacing and see which keeps the bird from "hitting" while still letting you see through. Keep the best spacing.

Measure: The dot spacing that still lets a viewer see through clearly.

"A warbler flew fifteen hundred miles and got fooled by a window in the last mile. Two dollars of dots could have saved its life."

Problem City rain rushes off hot pavement into storm drains, flooding creeks with dirty, warm water that harms the fish and bugs living there. · **Solution** A kid-designed rain garden — a planted dip that soaks up runoff, cleans it, and cools it before it reaches the creek.

Mission M5 Reaching Roots, M15 Biocentric Architecture · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A tilted tray comparing bare pavement runoff to a soil-and-plant rain garden.

1 • Identify

When rain falls on roads and roofs, it can't soak in. It rushes into drains, picking up oil and heat, then dumps into creeks all at once. The sudden flood of dirty water hurts the fish and insects downstream.

Research questions

1. Why doesn't rain soak into pavement?
2. How do plants and soil clean runoff water?
3. What happens to a creek during a big storm?

2 • Interview an expert

Who: A stormwater engineer, a landscape designer, a creek-watch volunteer

Ask them

1. How much water can a rain garden hold?
2. Which plants clean and soak up water best?
3. Where should a rain garden go?
4. How do you keep it from becoming a mud pit?
5. How does it help creek animals downstream?

3 • Create

Materials

- A baking tray
- Foil for "pavement"
- Soil and small plants or sponge
- A cup of dirty water
- Gravel
- Books to tilt the tray

Build steps

1. Tilt the tray and cover the top half with foil pavement.
2. Dig a dip at the bottom and fill it with soil and plants.
3. Pour dirty water down the pavement.
4. Watch the rain garden catch and clean it.
5. Label the parts: pavement, runoff, rain garden, plants, clean creek.

4 • Iterate & 5 • Communicate

Try deeper dips or more plants and compare how much water is caught and how clean it comes out. Keep the best design.

Measure: How much water the rain garden holds before overflowing.

"Every storm, our streets fire a flood of hot, oily water straight at a creek. What if the city could drink the rain instead?"

Problem City pollinators have nowhere to feed or rest because rooftops are bare, baking deserts of tar and gravel. ·
Solution A kid-designed green roof — a shallow garden on top of a building that gives bees and butterflies food and a cool place to stop.

Mission M15 Biocentric Architecture, M2 Exploding Seeds · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** Two
 "rooftops" — bare and planted — compared for heat and pollinator visits.

1 • Identify

City rooftops are usually empty and blazing hot. Bees and butterflies flying over the city find no food or shade up there. A rooftop garden turns dead space into a sky-high meadow full of flowers.

Research questions

1. Why are bare rooftops so hot?
2. How can plants grow on a roof without heavy soil?
3. Which flowers feed the most pollinators?

2 • Interview an expert

Who: A green-roof designer, a rooftop beekeeper, a building engineer

Ask them

1. How heavy can a roof garden safely be?
2. Which plants survive rooftop wind and sun?
3. How does a green roof cool the building?
4. How do pollinators find a roof so high up?
5. How do you water plants up there?

3 • Create

Materials

- Two shoebox lids as roofs
- Foil/gravel for the bare roof
- Soil and small plants or moss
- A thermometer or your hand
- Paper flowers and bees
- A lamp for sunshine

Build steps

1. Make one bare roof and one planted roof.
2. Shine the lamp on both for a few minutes.
3. Feel or measure which roof is cooler.
4. Add paper flowers and "fly" bees to the green roof.
5. Label the parts: roof, growing layer, plants, pollinators, cool zone.

4 • Iterate & 5 • Communicate

Add more plants or a thicker growing layer and check if the roof gets even cooler and holds more flowers. Keep the best mix.

Measure: The temperature difference between the bare and green roofs.

"Fly over any city and you'll see acres of empty, boiling rooftops. Now imagine them buzzing with bees. We're planting the sky."

Problem Treeless city blocks trap so much heat that birds, insects, and people all suffer on hot days. · **Solution** A kid-designed street-tree plan that adds shade to cool the block and give wildlife food and shelter.

Mission M12 Forest Elder, M15 Biocentric Architecture · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** Two model blocks — bare and tree-shaded — compared for temperature under a lamp.

1 • Identify

Pavement and buildings soak up sunshine and turn city blocks into ovens. Without trees, there's no shade or cool air, and birds and insects can't handle the heat. Street trees cool the block and feed wildlife at the same time.

Research questions

1. Why do treeless blocks get so hot?
2. How does a tree cool the air around it?
3. What wildlife depends on city street trees?

2 • Interview an expert

Who: An urban forester, a city planner, a wildlife biologist

Ask them

1. How much can one tree cool its surroundings?
2. Which trees survive best along streets?
3. How do trees help birds and bugs in the city?
4. What do city trees need to stay healthy?
5. How do you decide where to plant?

3 • Create

Materials

- Two shoebox-lid "blocks"
- Gray paper for pavement
- Craft-stick trees with paper leaves
- A lamp for sun
- A thermometer or your hand
- Tape

Build steps

1. Cover both blocks with pavement.
2. Add craft-stick trees to one block only.
3. Shine the lamp on both blocks for a few minutes.
4. Compare which block is cooler in the shade.
5. Label the parts: pavement, street trees, shade, wildlife, cool air.

4 • Iterate & 5 • Communicate

Add more trees or bigger canopies and see how much cooler the shaded block gets. Keep the layout that cools best.

Measure: The temperature under the trees versus the bare block.

"On a hot summer day, one street is an oven and the next is ten degrees cooler. The only difference? A row of trees. Let's plant the difference."

Problem Migrating monarch butterflies can't find the milkweed they need to lay eggs and refuel as they cross the city.

Solution A kid-designed "monarch waystation" garden of milkweed and nectar flowers planted in a median or park patch.

Mission M2 Exploding Seeds, M14 Seeds of Renewal · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A model garden strip with paper milkweed and flowers a monarch "visits" in sequence.

1 • Identify

Monarch butterflies travel thousands of miles and lay their eggs only on milkweed — the one plant their caterpillars can eat. Cities have paved over milkweed, so monarchs cross with no place to stop, feed, or lay eggs.

Research questions

1. Why can monarch caterpillars only eat milkweed?
2. What do adult monarchs drink for energy?
3. How far do monarchs travel each year?

2 • Interview an expert

Who: A monarch researcher, a native-plant gardener, a parks department planner

Ask them

1. How much milkweed does a waystation need?
2. Which nectar flowers do monarchs love?
3. When do monarchs pass through our area?
4. How do monarchs find a small garden?
5. How do you tag and track a monarch?

3 • Create

Materials

- A long box lid for a median
- Green paper milkweed
- Colorful paper nectar flowers
- A paper monarch
- Clay caterpillars
- Glue and markers

Build steps

1. Lay out a median strip garden.
2. Plant paper milkweed and nectar flowers together.
3. Add clay caterpillars on the milkweed leaves.
4. "Fly" the monarch to drink, then lay eggs on milkweed.
5. Label the parts: median, milkweed, nectar flowers, caterpillar, monarch.

4 • Iterate & 5 • Communicate

Try grouping milkweed in clumps versus spreading it out, and see which a monarch finds fastest. Keep the easiest-to-find layout.

Measure: How quickly a model monarch finds milkweed to lay eggs.

"A butterfly the weight of a paperclip is flying from Canada to Mexico — and it needs to find one specific plant in our whole city. Let's make sure it's here."

Problem Rain rushes off school roofs into storm drains, wasting clean water and dumping dirty runoff into creeks where wildlife lives. · **Solution** Kids set up a rain barrel that catches roof water and slowly waters a wildlife/pollinator garden instead of flooding the drain.

Mission M5 Reaching Roots, M15 Biocentric Architecture · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A model roof + downspout + barrel with a spigot that waters a “garden” tray.

1 • Identify

A single big roof sheds hundreds of gallons in one storm. That water blasts into drains carrying oil and litter to creeks, while gardens go thirsty. A rain barrel saves the water and lets it feed plants slowly.

Research questions

1. How much water comes off our roof in one storm?
2. Where does our storm-drain water end up?
3. What could that saved water grow?

2 • Interview an expert

Who: A stormwater engineer, a school facilities manager, a master gardener

Ask them

1. Where does our roof water go now?
2. How much does one storm shed?
3. What’s in “clean” runoff that hurts creeks?
4. Is a rain barrel safe and allowed at a school?
5. What plants would love that water?

3 • Create

Materials

- A jug or bottle (the barrel)
- Cardboard “roof” + straw downspout
- A tray “garden” with soil
- A spigot (tape a straw)
- Water
- Labels

Build steps

1. Build a slanted cardboard roof over a downspout.
2. Lead the downspout into the barrel jug.
3. Add a low spigot straw to the barrel.
4. Run the spigot into the garden tray.
5. Label: roof, downspout, barrel, spigot, garden.

4 • Iterate & 5 • Communicate

Pour “rain” on the roof and compare how much reaches the garden vs. spills to the “drain”; adjust the downspout to capture more.

Measure: The fraction of poured water captured in the barrel vs. lost to the drain.

“Watch two roofs in a rainstorm: one floods the creek, the other grows a butterfly garden. The only difference is a barrel.”

Problem City bats have lost the hollow trees and gaps they roost in, so they can't stay to eat the mosquitoes that swarm neighborhoods. · **Solution** A kid-designed bat box mounted high on a warm wall that gives bats a safe daytime roost.

Mission M11 Window to the Past, M13 Keystone Species · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A slim bat box tested to see if it stays warm and dark inside like bats need.

1 • Identify

Bats sleep by day in tight, warm, dark spaces and hunt bugs at night. Cities have torn down the old trees and buildings where they roosted. With nowhere to rest, bats leave, and mosquitoes multiply.

Research questions

1. Why do bats need warm, narrow roosts?
2. How many mosquitoes can a city bat eat?
3. Why did bats lose their city roosts?

2 • Interview an expert

Who: A bat conservation specialist, a pest-management expert, a city naturalist

Ask them

1. How high should a bat box go?
2. Why must the inside be warm and dark?
3. How do bats find a new roost box?
4. How do bats help control insects?
5. How do you know bats are using the box?

3 • Create

Materials

- A slim cardboard box
- Black paper
- A rough paper "landing" strip
- A thermometer or your hand
- A lamp for warmth
- Tape

Build steps

1. Make a tall, narrow box open at the bottom.
2. Line it dark and add a rough climbing strip inside.
3. Warm one side with a lamp like a sunny wall.
4. Check that inside stays warm and dark.
5. Label the parts: roost box, landing strip, dark chamber, warm wall, exit.

4 • Iterate & 5 • Communicate

Try painting it dark or facing it toward the "sun" and check which keeps the inside warmest. Keep the warmest design.

Measure: The inside temperature after warming in the "sun."

"One little brown bat eats hundreds of mosquitoes a night — for free. We just have to give it a place to sleep. Meet its new apartment."

Problem On hot city days, bees and butterflies can't find safe, shallow water and get too weak to keep pollinating.

Solution A kid-designed pollinator watering station with pebbles and shallow water so insects can drink without drowning.

Mission M4 Lucky Leaves, M10 Fragile Microhabitats · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A shallow dish with pebble "islands" tested so a floating "bee" can land and sip safely.

1 • Identify

Pollinators need water, not just flowers. But deep birdbaths and puddles can drown a tiny bee. On hot days with no safe drink nearby, bees and butterflies weaken and stop visiting flowers.

Research questions

1. Why do bees and butterflies need water?
2. Why is deep water dangerous for insects?
3. What makes a safe place for a bug to drink?

2 • Interview an expert

Who: A beekeeper, a butterfly gardener, an entomologist

Ask them

1. How do bees carry water back home?
2. What do butterflies get from "puddling"?
3. How shallow should insect water be?
4. Where should a water station go?
5. How do you keep the water clean?

3 • Create

Materials

- A shallow dish or lid
- Pebbles and marbles
- Water
- A cork or foam "bee"
- A dropper
- Sand

Build steps

1. Fill the dish with a thin layer of water.
2. Pile pebbles so their tops poke above the water.
3. Add a few sandy spots for butterflies.
4. Float or land the "bee" on a pebble to sip safely.
5. Label the parts: shallow water, pebble islands, landing spots, drinking bee, refill line.

4 • Iterate & 5 • Communicate

Adjust the water depth and pebble height so the "bee" always has a dry landing. Keep the safest, easiest-to-reach design.

Measure: How many safe landing spots stay dry above the water.

"A bee visited five hundred flowers today and now it's dying of thirst three feet from a birdbath it can't safely use. Let's set the table with a drink."

Problem Winter road salt washes off city streets into creeks, making the water too salty for fish, frogs, and insects to survive. · **Solution** A kid-designed "smart salting" plan and a salt-catching buffer strip that keeps salty runoff out of the water.

Mission M5 Reaching Roots, M9 Research Platform · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A ramp test comparing salty runoff reaching a "creek" with and without a plant buffer.

1 • Identify

Cities spread salt to melt ice, but when snow melts, the salt runs into creeks. Too much salt poisons freshwater animals that can't handle it. A little goes a long way, and buffers can catch what's spilled.

Research questions

1. Why does salt melt ice on roads?
2. How does too much salt hurt creek animals?
3. How can plants and soil catch runoff?

2 • Interview an expert

Who: A watershed scientist, a public-works road crew, a stream ecologist

Ask them

1. How do you measure salt in a creek?
2. How much salt is too much for fish?
3. How can cities use less salt safely?
4. Do buffer strips really catch runoff?
5. When are salt levels the worst?

3 • Create

Materials

- A baking tray
- Foil for the road
- Soil and plants or sponge
- Salty water (colored)
- A clear cup as the creek
- Books to tilt

Build steps

1. Tilt the tray with a foil road at the top.
2. Set a cup "creek" at the bottom.
3. Pour salty water down the bare road into the creek.
4. Add a plant buffer strip and pour again.
5. Label the parts: road, salty runoff, buffer strip, creek, filtered water.

4 • Iterate & 5 • Communicate

Make the buffer wider or thicker and see how much less salty water reaches the creek. Keep the buffer that catches the most.

Measure: How much runoff the buffer holds back from the creek.

"The salt that kept our sidewalk safe in January is quietly poisoning a creek in March. There's a smarter way — and we built it."

Problem Cities have almost no small ponds, so frogs and dragonflies have nowhere to breed and vanish from neighborhoods. · **Solution** Kids build a safe “pocket pond” in a tub — shallow, planted, with an escape ramp — so frogs, dragonflies, and birds have water in the city.

Mission M10 Fragile Microhabitats, M9 Research Platform · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A tub pocket-pond model with a shallow shelf, plants, a pebble ramp, and paper wildlife.

1 • Identify

Frogs and dragonflies need still water to lay eggs, but cities pave over ponds and puddles. A small container pond — even a tub — can bring them back, if it’s shallow with a way for animals to climb out.

Research questions

1. What wildlife would use a small city pond?
2. Why does a pond need a shallow edge and a ramp?
3. Where is a safe spot for one at school?

2 • Interview an expert

Who: An amphibian biologist, a naturalist, a landscape designer

Ask them

1. What makes a pond safe for frogs, not mosquitoes?
2. Why is a shallow shelf important?
3. How do animals get out so they don’t drown?
4. What plants keep the water healthy?
5. Where should we NOT put a pond?

3 • Create

Materials

- A tub or basin
- Pebbles / stones
- Small plants or greenery
- A ramp (rock pile or board)
- Water
- Paper frogs / dragonflies

Build steps

1. Set the tub and build a shallow pebble shelf on one side.
2. Add a stone ramp from bottom to rim.
3. Place plants in the shallow zone.
4. Fill with water; add paper wildlife.
5. Label: deep zone, shallow shelf, ramp, plants.

4 • Iterate & 5 • Communicate

Test the ramp by “walking” a toy frog out; make the slope gentle enough that nothing gets stuck.

Measure: The gentlest ramp slope that still lets a toy animal climb out of the water.

“A frog pond doesn’t need a backyard — it needs a tub, some stones, and a way out. We’ll show you how a bucket becomes a habitat.”

Problem In late summer and early fall, farm pollinators run out of blooms and go hungry before winter. · **Solution** A kid-designed "late-bloomer strip" of fall-flowering plants that feeds bees through the hungry gap.

Mission M4 Lucky Leaves, M14 Seeds of Renewal · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A model bloom calendar strip showing a food gap and the fall flowers that fill it.

1 • Identify

Bees need flowers all season, but on many farms the big blooms finish by midsummer. Then comes a "hungry gap" with little to eat, right before bees must store food for winter. Late-blooming plants can bridge it.

Research questions

1. What is the pollinator "hungry gap"?
2. Which flowers bloom late in the season?
3. Why do bees need food before winter?

2 • Interview an expert

Who: A beekeeper, a farm agronomist, a native-plant grower

Ask them

1. When does forage run short on farms?
2. Which fall flowers feed bees best?
3. How much food do bees store for winter?
4. Where can farmers fit flower strips?
5. How do you know bees are well-fed?

3 • Create

Materials

- A long strip of paper for a calendar
- Colored paper flowers
- Markers
- A paper bee
- Glue
- A ruler

Build steps

1. Draw the months across the calendar strip.
2. Place spring and summer flowers where they bloom.
3. Mark the empty "hungry gap" in late summer.
4. Add fall flowers to fill the gap and "fly" the bee across.
5. Label the parts: bloom calendar, hungry gap, late bloomers, bee, winter store.

4 • Iterate & 5 • Communicate

Add different fall flowers and check whether the bee can find food every week with no gaps. Keep the mix with no empty weeks.

Measure: The number of weeks with at least one flower in bloom.

"Picture a grocery store that closes every August — right when you need to stock up for winter. That's a bee's life on many farms. Let's keep the shelves full."

Problem The orchard's fruit trees aren't getting fully pollinated because there aren't enough native bees nearby to visit every blossom. · **Solution** A kid-designed native-bee habitat of nesting tunnels and early blooms placed beside the orchard.

Mission M6 Leafcutter Frenzy, M2 Exploding Seeds · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A bundle of paper tubes as a native-bee nest tested for the right tunnel size.

1 • Identify

Apple and cherry trees need bees to move pollen between blossoms so they grow fruit. Native bees like mason bees are great early-spring pollinators. Without nesting spots nearby, few show up and much of the fruit never forms.

Research questions

1. How does a bee help a blossom become fruit?
2. Why are native bees good early pollinators?
3. Where do native bees make their nests?

2 • Interview an expert

Who: An orchard grower, a native-bee researcher, an extension agent

Ask them

1. How many bees does an orchard need?
2. What size tunnels do mason bees like?
3. When do fruit trees need pollinators most?
4. How do native bees differ from honeybees?
5. How do you keep a bee nest healthy?

3 • Create

Materials

- Paper rolled into tubes of different widths
- A tin can or box to hold them
- Blossom pictures
- A paper bee
- Tape
- A ruler

Build steps

1. Roll paper into tubes of a few different widths.
2. Bundle them inside a can, all facing out.
3. Place the nest beside a paper orchard.
4. Move the bee from blossoms to a snug tunnel.
5. Label the parts: nest bundle, tunnels, orchard, blossoms, native bee.

4 • Iterate & 5 • Communicate

Test which tube width the "bee" fits snugly and set more of those. Keep the sizes native bees actually use.

Measure: Which tunnel width fits the model bee best.

"No bees, no apples — it's that simple. Half these blossoms will never become fruit unless someone shows up to work. Meet the orchard's tiny crew."

Problem Hawks that keep mice off the farm have nowhere to perch and hunt over the open fields. · **Solution** A kid-designed hunting perch — a tall post with a crossbar — that gives raptors a lookout to control rodents naturally.

Mission M12 Forest Elder, M13 Keystone Species · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A tall perch post tested to stand steady in "wind" while a paper hawk sits on top.

1 • Identify

Hawks and kestrels hunt mice and voles that eat crops. But they need a high perch to spot prey in open fields. With no tall trees or posts, the raptors move on and rodents multiply, damaging the harvest.

Research questions

1. How do hawks use perches to hunt?
2. What happens to crops when rodents increase?
3. Why is natural pest control better than poison?

2 • Interview an expert

Who: A raptor biologist, a farmer, an integrated pest management (IPM) advisor

Ask them

1. How tall should a hunting perch be?
2. How many mice can a kestrel eat in a day?
3. Where in a field do perches work best?
4. Why is poison risky for hawks?
5. How do you know raptors are using a perch?

3 • Create

Materials

- A dowel or pencil for the post
- A craft stick crossbar
- A clay base
- A paper hawk
- A fan or your breath
- Tape

Build steps

1. Stand the post upright in a clay base.
2. Fix a crossbar near the top for a perch.
3. Set the paper hawk on the crossbar.
4. Blow "wind" and see if it stays standing.
5. Label the parts: post, crossbar, base, hawk, field.

4 • Iterate & 5 • Communicate

Widen the base or shorten the post until it stands firm in strong wind with the hawk on top. Test each version the same way.

Measure: The wind strength the perch survives without tipping.

"One kestrel on a fencepost can out-hunt a bucket of poison — and it works for free, all season. The farm just needs to give it a chair."

Problem In the pasture, manure piles up because too few dung beetles are around to bury it and recycle its nutrients into the soil. · **Solution** A kid-designed "beetle-friendly pasture" plan that protects dung beetles so they clean manure and feed the grass.

Mission M3 Flip the Rock, M6 Leafcutter Frenzy · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** Two soil trays comparing how fast "manure" disappears with and without beetle helpers.

1 • Identify

Dung beetles roll and bury animal manure, turning waste into rich soil and cutting down on flies. On pastures where beetles are harmed, manure sits and sours, wasting nutrients and breeding pests.

Research questions

1. What does a dung beetle do with manure?
2. How does buried manure help the soil?
3. What harms dung beetles on farms?

2 • Interview an expert

Who: A pasture ecologist, a cattle farmer, an entomologist

Ask them

1. How much manure can beetles bury?
2. How do buried nutrients help grass grow?
3. What farm chemicals hurt dung beetles?
4. How do beetles cut down farm flies?
5. How do you attract more beetles?

3 • Create

Materials

- Two soil trays
- Clay or mud "manure"
- Craft "beetles"
- A spray bottle
- Grass seed or paper grass
- A marker

Build steps

1. Place equal "manure" lumps on both trays.
2. Add beetles that "bury" manure in one tray.
3. Leave the other tray with no beetles.
4. Compare how fast manure disappears and grass grows.
5. Label the parts: pasture, manure, dung beetles, buried nutrients, new grass.

4 • Iterate & 5 • Communicate

Add more beetles or moist soil and see how much faster manure is buried. Keep the setup that recycles fastest.

Measure: How much "manure" is buried after a set time.

"Nobody claps for the dung beetle. But without this tiny recycler, the pasture would drown in waste and flies. Meet the farm's unpaid cleanup crew."

Problem The farm's bee hotel is spreading disease and mites because its tunnels can't be cleaned and get reused year after year. · **Solution** A kid-designed bee hotel with removable, cleanable liners so solitary bees stay healthy.

Mission M6 Leafcutter Frenzy, M10 Fragile Microhabitats · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A bee hotel with paper-liner tubes that can slide out for cleaning.

1 • Identify

Bee hotels give solitary bees a place to nest, but if the tunnels are never cleaned, mites and mold build up and make the bees sick. A hotel you can open and clean keeps the next generation healthy.

Research questions

1. Why do solitary bees need nesting tunnels?
2. How does a dirty bee hotel spread disease?
3. Why must tunnels be cleaned each year?

2 • Interview an expert

Who: A native-bee specialist, a beekeeper, a farm educator

Ask them

1. How often should a bee hotel be cleaned?
2. What pests build up in old tunnels?
3. What tunnel size do solitary bees prefer?
4. How do you clean without harming bees?
5. Where should a bee hotel be placed?

3 • Create

Materials

- A small box or can
- Paper rolled into removable liner tubes
- A pencil for sizing
- Tape
- Markers
- A paper bee

Build steps

1. Roll paper liners that slide into the box snugly.
2. Fill the box with the removable tubes.
3. Show a bee "nesting" in a tube.
4. Slide the liner out to clean and replace it.
5. Label the parts: hotel box, removable liners, tunnels, nesting bee, clean-out.

4 • Iterate & 5 • Communicate

Test whether liners slide out easily and fit snugly, then adjust the size. Keep the design that's easiest to clean.

Measure: How quickly you can remove and replace all the liners.

"We built the bees a hotel — and forgot to ever clean the rooms. Now it's making them sick. Here's a hotel with a housekeeping plan."

Problem The farm's old wildflower meadow is gone, and there are no local seeds left to bring back the plants pollinators depend on. · **Solution** A kid-designed community seed bank that collects, dries, and stores native seeds to replant the meadow.

Mission M14 Seeds of Renewal, M2 Exploding Seeds · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A labeled seed-storage tray tested for keeping seeds dry so they still sprout.

1 • Identify

Native wildflower meadows feed pollinators, but when a meadow is plowed or paved, its special local seeds can be lost forever. Saving and storing seeds now means the meadow — and its bees and butterflies — can come back later.

Research questions

1. Why are local native seeds so important?
2. What do seeds need to stay alive in storage?
3. How does a meadow help pollinators?

2 • Interview an expert

Who: A seed-bank curator, a restoration ecologist, a native-plant gardener

Ask them

1. How do you collect seeds without harming plants?
2. How must seeds be dried and stored?
3. How long can seeds stay alive?
4. When is the best time to gather seeds?
5. How do you replant a meadow from seed?

3 • Create

Materials

- A tray with small cups or an egg carton
- Real or paper seeds
- Labels and markers
- Paper towels for drying
- A silica packet or rice
- A lid

Build steps

1. Sort seeds into labeled cups by plant type.
2. Dry them on paper towels first.
3. Add rice or a silica packet to keep them dry.
4. Cover and store the tray in a cool, dark spot.
5. Label the parts: seed bank, sorted seeds, labels, drying agent, storage lid.

4 • Iterate & 5 • Communicate

Compare seeds kept dry versus damp and see which stay ready to sprout. Keep the storage method that works best.

Measure: How dry the stored seeds stay over several days.

"This jar holds a meadow. Inside are the last seeds of wildflowers that fed bees here for a hundred years. We're saving tomorrow's spring in a cup."

Problem A new invasive plant or bug can spread across the farm before anyone notices, crowding out native species pollinators need. · **Solution** A kid-designed "spotter card" and mapping plan that helps farmers catch and report invaders early.

Mission M1 Drone Survey, M11 Window to the Past · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A field map grid where "spotters" mark invasive finds early before they spread.

1 • Identify

Invasive species are plants or animals from elsewhere that spread fast and push out native life. If caught early, a few can be removed easily. If missed, they take over and starve the native plants and pollinators.

Research questions

1. What makes a species "invasive"?
2. Why is finding invaders early so important?
3. How do invasive species harm native wildlife?

2 • Interview an expert

Who: An extension agent, an invasive-species biologist, a farmer

Ask them

1. Which invaders threaten our area now?
2. How do you tell a native from an invader?
3. How do you report a new sighting?
4. How fast can an invader spread?
5. What's the best way to remove one early?

3 • Create

Materials

- A large paper grid map
- Colored stickers or markers
- Cards with "invader" pictures
- A ruler
- A pencil
- Tape

Build steps

1. Draw a grid map of the farm fields.
2. Make spotter cards showing invaders to watch for.
3. "Walk" the fields and mark any finds on the grid.
4. Circle early finds and show the removal plan.
5. Label the parts: field map, grid squares, spotter cards, sightings, removal zone.

4 • Iterate & 5 • Communicate

Try checking the map more often and see how much smaller the invasion stays. Keep the routine that catches invaders earliest.

Measure: How many grid squares get invaded before it's caught.

"One weed today, a whole field lost by next summer. The difference is whether someone spotted it in time. We're training the farm's early-warning team."

Problem Rats and voles damage the farm's stored grain, and poison meant for them also kills the owls that could control them. · **Solution** A kid-designed barn owl nest box that invites a family of owls to hunt rodents naturally. **Mission** M13 Keystone Species, M12 Forest Elder · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A nest box tested for the right entrance size and a dark, safe interior for owls.

1 • Identify

Barn owls are silent night hunters that eat huge numbers of rats and voles. But rodent poison moves up the food chain and kills the owls too. Give owls a safe nest box, and one family can outwork any poison.

Research questions

1. How many rodents does a barn owl family eat?
2. Why does rat poison also harm owls?
3. What kind of nest do barn owls need?

2 • Interview an expert

Who: A raptor biologist, a farmer, a wildlife rehabber

Ask them

1. Where should an owl box be placed?
2. What size entrance do barn owls need?
3. How do owls find a new nest box?
4. Why is poison dangerous up the food chain?
5. How do you know owls have moved in?

3 • Create

Materials

- A cardboard box
- A round cutter for the entrance
- Black paper
- Wood shavings or paper bedding
- Tape
- A paper owl

Build steps

1. Build a box big enough for an owl family.
2. Cut a round entrance hole to the right size.
3. Line it dark with soft bedding inside.
4. Show the owl entering and roosting.
5. Label the parts: nest box, entrance hole, dark chamber, bedding, owl.

4 • Iterate & 5 • Communicate

Test different entrance sizes so the owl fits but bigger predators can't. Keep the safest size.

Measure: Whether the "owl" fits the entrance while larger shapes don't.

"A single barn owl family eats a thousand rats a season — no poison, no cost. All they ask for is a dark box up in the barn. Let's hang the welcome sign."

Problem The farm pond's steep, bare banks give frogs and toads nowhere to lay eggs or hide, so their numbers crash.

Solution A kid-designed gentle, planted pond edge that gives amphibians a safe place to breed and shelter.

Mission M10 Fragile Microhabitats, M8 Tangled · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A pond model comparing a steep bare bank to a gentle planted shelf where "frogs" can rest.

1 • Identify

Frogs and toads lay eggs in still shallow water and need plants and cover to hide from predators. A pond with steep, bare, mowed banks offers no safe shallows. Without good edges, tadpoles don't survive and the frogs fade away.

Research questions

1. Why do frogs need shallow, planted water?
2. What eats tadpoles, and how do plants help?
3. Why are steep bare banks bad for amphibians?

2 • Interview an expert

Who: A herpetologist, a pond-management specialist, a farmer

Ask them

1. How shallow should a frog-friendly edge be?
2. Which plants help tadpoles hide and grow?
3. When do frogs lay their eggs?
4. How do frogs help control farm pests?
5. How do you tell if a pond is healthy?

3 • Create

Materials

- A plastic tub
- Soil and gravel
- Small plants or moss
- Water
- Clay "frogs" and "eggs"
- A ruler

Build steps

1. Build a steep bare bank on one side of the tub.
2. Build a gentle, planted shelf on the other side.
3. Add water to make a shallow zone on the shelf.
4. Place frogs and eggs where they can hide safely.
5. Label the parts: pond, steep bank, gentle shelf, plants, tadpole zone.

4 • Iterate & 5 • Communicate

Try different bank slopes and amounts of plants and see which gives frogs the most safe, shallow space. Keep the best edge.

Measure: How much shallow, planted area the edge provides.

"A frog is looking for the perfect spot to raise its tadpoles — and finding only a steep, slippery wall of mud. Let's build the nursery it needs."

Problem Big open crop fields have no cover in the middle, so the beetles and spiders that eat crop pests can't live there — and pests take over. · **Solution** Kids design a "beetle bank" — a raised, grassy strip through the field — giving pest-eating bugs a home base right next to the crops.

Mission M6 Leafcutter Frenzy, M13 Keystone Species · **Connects** BIOBUZZ ★, BIOCORE · **Prototype** A field model with a raised grassy bank strip and beetles that "patrol" out into the crops.

1 • Identify

Ground beetles and spiders eat huge numbers of crop pests for free, but they can't cross a wide bare field. A raised, grassy ridge running through the field gives them shelter and a launch pad right where the pests are.

Research questions

1. Which good bugs eat crop pests?
2. Why can't they cross a big bare field?
3. What kind of strip would shelter them?

2 • Interview an expert

Who: A farmer, an entomologist, an IPM (pest-management) advisor

Ask them

1. Which beneficial bugs live on your farm?
2. How far will a ground beetle travel from cover?
3. What grasses make good beetle habitat?
4. Does a beetle bank really cut pesticide use?
5. What's the hardest part of adding one?

3 • Create

Materials

- A tray "field" with soil
- A strip of grass / green scrub (raised bank)
- Small paper beetles / spiders
- "Crop" rows (toothpicks / paper)
- "Pest" markers
- Labels

Build steps

1. Lay out crop rows across the tray field.
2. Build a raised grassy bank strip down the middle.
3. Place beetles in the bank.
4. Scatter "pests" through the crop.
5. Label: crop rows, beetle bank, cover, patrol range.

4 • Iterate & 5 • Communicate

Move the bank closer/farther from the rows and see how much of the field the beetles can "reach"; find the spacing that covers the most crop.

Measure: How far from the bank the beetles can patrol before pests go uneaten.

"The best pest control on a farm has six legs and works for free — if you give it one grassy ridge to call home."